Math 344, F19 **Name:** Answer

Quiz 13

<u>**Instructions:**</u> **There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (10 points) Find an orthogonal matrix $\mathbf{Q}$ that diagonalizes $\mathbf{A} = \begin{bmatrix} -2 & 6 & 0 \\ 6 & 7 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, i.e., $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathbf{T}}$.

$$\det(A-\lambda I) = \begin{vmatrix} -2-\lambda & 6 & 0 \\ 6 & 7-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix}$$

$$= [(-2-\lambda)(7-\lambda)-36](3-\lambda)$$

$$= (\lambda^2-5\lambda-50)(3-\lambda)$$

$$= (\lambda-10)(\lambda+5)(3-\lambda)$$

$$\Rightarrow \lambda_1 = 10,\ \lambda_2 = -5,\ \lambda_3 = 3 \Rightarrow \Lambda = \begin{bmatrix} 10 & & \\ & -5 & \\ & & 3 \end{bmatrix}$$

For $\lambda_1 = 10$, $(A-10I)X_1 = 0$

$$\begin{bmatrix} -12 & 6 & \\ 6 & -3 & \\ & & -7 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{matrix} -12x_1 + 6x_2 = 0 \\ -7x_3 = 0 \end{matrix} \Rightarrow X = c_1\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

let $c_1 = \sqrt{5}$, then $X_1 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix} \to \|X_1\| = 1$

For $\lambda_2 = -5$, $(A+5I)X_2 = 0$

$$\begin{bmatrix} 3 & 6 & \\ 6 & 12 & \\ & & 8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{matrix} 3x_1 + 6x_2 = 0 \\ 8x_3 = 0 \end{matrix} \Rightarrow X = c_2\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

let $c_2 = \sqrt{5}$, then $X_2 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix} \to \|X_2\| = 1$

For $\lambda_3 = 3$, $(A-3I)X_3 = 0$

$$\begin{bmatrix} -5 & 6 & \\ 6 & 4 & \\ & & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{matrix} -5x_1 + 6x_2 = 0 \\ 6x_1 + 4x_2 = 0 \end{matrix} \Rightarrow x_1 = x_2 = 0 \Rightarrow X_3 = c_3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

let $c_3 = 1$, then $X_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \to \|X_3\| = 1$

thus $Q = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} & 0 \\ 2/\sqrt{5} & 1/\sqrt{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}$